**Respiratory Assessment**

EMR — **Titrate Oxygen**
To maintain SpO2 <94%

Dilute nebulized Salbutamol and Combivent with Normal Saline or Sterile Water to a maximum volume of 5ml

## Mild Asthma

EMT — **Nebulized Salbutamol**

| Adult | Pediatric |
|---|---|
| 5mg Neb | **< 5 years** 2.5mg Neb |
| | **> 5 years** 5mg Neb |
| Repeat after 15 minutes if required. | |

## Moderate Asthma

Administer treatment **as per mild asthma**, and:

| | Adult | Pediatric |
|---|---|---|
| EMT — **Nebulized Combivent** | 2.5mg Salbutamol 0.5mg Ipratropium Bromide | Adult Only |
| EMT — **Hydrocortisone IM** / EMT-A — **Hydrocortisone IV** | 100mg IV/IM | **< 1 year** 25mg IV/IM |
| | | **1 - 5 years** 50mg IV/IM |
| | | **> 5 years** 100mg IV/IM |

## Severe Asthma

Administer treatment **as per moderate asthma**, and:

Consider ALS Back-Up

AP — **Magnesium Sulphate**

| Adult | Pediatric |
|---|---|
| 2g IV/IM | 40mg/kg IV/IM (max 2mg) |
| Give IV infusion over 20minutes | |

## Life Threatening Asthma

Administer treatment **as per Severe asthma**, and:

| | Adult | Pediatric |
|---|---|---|
| AP — **Adrenaline IM** | 0.3-0.5mg | 0.15-0.3mg |
| AP — **Adrenaline IV Infusion** | 2-10**mcg**/min | 0.1-1**mcg**/kg/min |

## Patient Care Goals

1. Conduct a rapid assessment focusing on airway, breathing, circulation, and disability (ABCD).
2. Perform respiratory assessment to identify severity.
3. Assure adequate oxygenation and ventilation.
4. Recognize impending respiratory failure.
5. Promptly identify and intervene for patients who require escalation of therapy.
6. Alleviate respiratory distress.

## Patient Inclusion & Exclusion Criteria

**Inclusion Criteria**

- Patients with mild, moderate, severe, or life-threatening asthma based on respiratory assessment findings.

**Exclusion Criteria**

- Patients presenting with difficulty of breathing due to another cause (e.g. COPD, bronchiolitis, pulmonary edema).

## Key Patient Assessment Points

1. **Patient Assessment**
   - Conduct a primary and secondary survey as per Universal Care CPG.
2. **Patient Monitoring**
   - Attach SpO2 monitor and ECG to all patients to continuously monitor oxygen saturation and cardiac rhythm throughout the assessment and treatment process.
   - Add NC-EtCO2 (via Nasal Cannula) monitoring for all moderate to severe and life-threatening cases.
3. **Respiratory Assessment**
   - Perform a respiratory assessment, including:
     - Auscultation: Listen for wheezes, silent chest, or diminished breath sounds.
     - Observation: Assess respiratory rate, effort, and use of accessory muscles.
     - Palpation: Evaluate the chest for deformities or tenderness.
     - Assess for signs of severe or life-threatening asthma: silent chest, Glasgow Coma Scale (GCS) <14, blood pressure <90/60 mmHg, or very slow/gasping breathing.
4. **Severity Assessment**
   - Assess severity of Asthma

| | Mild | Moderate | Severe* | Life-threatening* |
|---|---|---|---|---|
| **Talks In** | Sentences | Phrases | Words | Unable to speak |
| **Level of consciousness** | Conscious | Conscious | Agitated/distressed | Drowsy, collapsed |
| **Physical exhaustion** | No | No | Visibly breathless; increased work of breathing | Exhausted |
| **Accessory muscle use** | Nil | Mild accessory muscle use | Moderate accessory muscle use | Severe accessory muscle use, or minimal due to tiring |
| **Pulse rate** | Normal | Tachycardia | Tachycardia | May have tachycardia or pre-arrest bradycardia |
| **Respiratory rate** | May be normal or raised | Tachypnoea | May have tachypnoea | Poor respiratory effort; may be apnoeic |
| **Wheeze** | Variable volume, usually only expiratory | Moderate to loud; may have some inspiratory and expiratory | Due to poor air movement may be low volume (wheeze severity is poor predictor of severity) | Due to poor air movement may be low volume (wheeze severity is poor predictor of severity) |
| **Oximetry on presentation** | Normal, SpO2 usually >95% in air | Normal, SpO2 usually 92-95% in air | SpO2 90-94% in air | Hypoxic SpO2 <90% in air |

| | | | | |
|---|---|---|---|---|
| **Central cyanosis** | No | No | May be present | Present |
| **Posture** | - | - | Patient seated upright, unable to lie supine, pursed-lip breathing | Patient seated upright, unable to lie supine, pursed-lip breathing, hyper-inflated thorax |

- ***For severe or life-threatening asthma, patients only need to exhibit any of:**
  - **Severe:** Speaking words only or visibly breathless.
  - **Life-threatening:** Drowsy, collapsed, or exhausted or cyanotic or poor respiratory effort or SpO2 <90%.

5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

## Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
2. **Oxygen Therapy**
   - If a nebulizer is not available, and SpO2 <94%, administer oxygen therapy via a non-rebreather mask (NRB) to maintain SpO2 between 94-98%.
2. **Mild Asthma**
   - **Administer Nebulized Salbutamol:**
     - Adult: Salbutamol 5mg Neb.
     - Pediatric (<5 years): 2.5mg Neb.
     - Pediatric (>5 years): 5mg Neb.
     - Repeat after 15 minutes if required.
     - Dilute nebulized Salbutamol with Normal Saline or Sterile Water to a maximum of 5ml.
3. **Moderate Asthma**
   - Administer treatment as per mild asthma, and:
   - **Administer Nebulized Combivent**:
     - Adults: 0.5mg Ipratropium Bromide and 2.5mg Salbutamol, Nebulized.
       - May be repeated after 20minutes if required. Maximum 3 doses.
     - Pediatric: Not recommended for pediatric administration.
   - **Administer Hydrocortisone:**
     - Adults: Hydrocortisone 100mg IV/IM.
     - Pediatric (<1 year): 25mg IV/IM.
     - Pediatric (1-5 years): 50mg IV/IM.
     - Pediatric (>5 years): 100mg IV/IM.
4. **Severe Asthma**
   - Administer treatment as per moderate asthma, and:
   - **Request ALS Back-Up.**
   - **Administer Magnesium Sulphate**
     - Adults: 2g IV/IM. Administer as an infusion over 20 minutes.
     - Pediatric: 40mg/kg IV/IM, maximum 2g, Administer as an infusion over 20 minutes.
5. **Life-Threatening Asthma**
   - Administer treatment as per severe asthma, and:
   - **Administer Adrenaline IM**
     - Adults: IM 0.3-0.5mg
     - Pediatric: IM 0.15-0.3mg.
   - **Administer IV Adrenaline**
     - Adults: IV infusion 2-10mcg/min.
     - Pediatric: IV infusion 0.1-1mcg/kg/min.
6. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

### Patient Safety Considerations

- Use a syringe driver for IV infusions when available.
- It is not recommended to administer IV adrenaline infusions without a syringe driver to adults and pediatrics.

### Clinical Insights & Additional Guidance

- Call for back-up early.
  - Severe and life-threatening asthmatics are at high risk of deterioration and respiratory arrest.
- Hydrocortisone
  - Hydrocortisone is preferred to be given IV in adults with severe asthma.
  - Hydrocortisone in severe asthma can be increased to:
    - Adults: 200mg.
    - Pediatrics: 4mg/kg.
  - If a patient has deteriorated into the severe or life-threatening category, and hydrocortisone has already been administered, it is not a priority to administer a "top-up" dose.
- IV Adrenaline:
  - IV adrenaline should be administered via a syringe driver where possible.
  - Careful patient assessment should be undertaken prior to increasing the IV adrenaline administration rate to determine the difference between worsening asthma and the effects of IV adrenaline.
  - IV Adrenaline administration requires careful monitoring, especially in pediatrics, using a syringe driver and assess for worsening asthma vs. adrenaline effects.

### Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Presenting symptoms (e.g., wheezing, respiratory distress), initial respiratory status.
- Injuries / Information: Relevant history (e.g., asthma triggers, allergies).
- Signs:
  - Full vital signs including pulse, respiratory rate, SpO2, EtCO2.
  - Two sets of vital signs unless critical patient condition precludes.
  - Respiratory assessment findings (e.g., wheeze, accessory muscle use).
- Treatment:
  - Oxygen administration and nebulized medication doses.
  - All interventions: Salbutamol, Combivent, Hydrocortisone, Magnesium Sulphate, Adrenaline, and response.
  - Justification if interventions were withheld.
- Allergies: Documented allergies and drug interactions.
- Medications: All pre-existing medications and any administered in the field.
- Background: Relevant medical history, comorbidities, and recent procedures.
- Other:
  - Communication with Telemedicine (time and instructions).
  - Response to interventions.
  - Consideration of reversible causes and actions taken.

### Key Performance Indicators

1. Document respiratory assessment findings.
2. Administer treatment (e.g. salbutamol) for moderate to life threatening cases within 10 minutes of EMS arrival.

### References

1. Ambulance Tasmania, 2023. *Clinical practice guidelines for ambulance services*. Available at: https://www.ambulance.tas.gov.au.
2. Ambulance Victoria, 2024. *Clinical practice guidelines*. Available at: https://www.ambulance.vic.gov.au.

3. British Columbia Emergency Health Services, 2024. *BCEHS treatment guidelines*. Available at: https://handbook.bcehs.ca.
4. East of England Ambulance Service NHS Trust, 2023. *Clinical guidelines*. Available at: https://www.eastamb.nhs.uk.
5. National Association of EMS Officials (NAEMSO), 2022. *Model EMS clinical guidelines*. Available at: https://naemso.org.
6. NSW Ambulance, 2023. *Clinical practice guidelines*. Available at: https://www.ambulance.nsw.gov.au.
7. Queensland Ambulance Service, 2024. *Clinical practice guidelines*. Available at: https://www.ambulance.qld.gov.au.
8. St John Ambulance Western Australia, 2023. *Clinical practice guidelines*. Available at: https://clinical.stjohnwa.com.au.
9. St John New Zealand, 2024. *Clinical practice guidelines*. Available at: https://www.stjohn.org.nz.
10. Welsh Ambulance Services NHS Trust, 2023. *Clinical guidelines*. Available at: https://www.ambulance.wales.nhs.uk.